

AI-Based Detection of Jaywalking Violations in Davao City

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Abstract – Ensuring pedestrian safety is one of the major responsibilities of growing urban cities like Davao City, where increasing traffic volume and road expansion have led to more frequent jaywalking accidents. Despite the existing ordinances, common enforcement remains inefficient due to its high manpower demand, limited coverage, and lack of continuous monitoring. This study offered an AI-based detection system to automatically identify jaywalking violations using real-time video analysis from CCTV cameras. The system integrates the YOLOv8 model to detect pedestrians and determine if crossings occur outside designated lanes, with an automated logging feature that records timestamps and visual evidence of each violation. Model performance was evaluated using precision, recall, F1-score, and accuracy, comparing a custom-trained model with pre-trained YOLOv8 model. Results showed that the developed model has strong accuracy and reliability in detecting jaywalking incidents under different urban conditions. Through automation, the system aims to assist Davao City's local government in traffic enforcement, reducing human workload, and promoting safer road behaviour. This project demonstrates how artificial intelligence and computer vision can support smarter, data-driven solutions for urban safety management.

Keywords – Jaywalking Detection, Artificial Intelligence, Computer Vision, YOLOv8, Traffic Safety, Davao City

I. INTRODUCTION

A. Background of the Topic

One of the essential duties of any smart and modern cities is the safety and protection of the pedestrians. By providing safe walking areas and efficient travel systems, they ensure that people crossing on the roads and streets are safe. However, as cities develop and grow, the expansion on roads to reduce congestion becomes a necessity [1]. This results to an increase in interaction between vehicles and pedestrians which can be a safety concern for the people. In addition, the expansion resulted to an increase of jaywalking incidents due to lack of crosswalks and to save time [2].

The term jaywalking refers to the pedestrians violating traffic laws which were designed to protect both the walkers and drivers [2]. This behaviour is a major safety concern in modern cities specifically in traffic environments with intense pedestrian

activity since it leads to accidents or disrupts traffic flow [3]. Thus, developing an effective strategy to mitigate jaywalking is one of the primary goals for city governments.

This problem is observed in the Philippines especially in some specific areas in Davao City. In 2018, over 40,800 jaywalkers were caught which is 38% of the total traffic violators in that year [4]. Although there was a 64% decrease in jaywalking violations in 2019 which reflects to an improved discipline and effective enforcement [5], the issue in jaywalking persists. As noted by traffic enforcers, jaywalking remains a significant safety and traffic concern in which traditional and manual enforcement method is not a complete solution anymore. It is manpower-intensive, costly, and cannot provide 24/7 monitoring in high-risk areas [6]. Using the power and concepts of Artificial Intelligence (AI) tools, this article introduces a system that would help solve the limitations of manual enforcement by utilizing existing CCTV cameras to automate the detection of violations and reduce the manpower burden on traffic enforcers.

Different countries implemented this type of system which effectively reduce the jaywalking incidents in their area. In 2018, China launched an AI system that uses facial recognition on CCTV to capture jaywalkers' images and display them on LED screens for public shaming [7]. In addition, they will receive a fine sent through SMS. Implementing this over 10 months resulted to a drop from 200,000+ annual offenders down to 13,930 [8]. If this type of system is applied in the Philippines specifically in Davao City, there is a high chance that jaywalking incidents occurring in the city will be reduced which will decrease road traffic accidents.

B. Objectives

This study aims to create an automated jaywalking detection system for Davao City using real-time video analysis. The primary objectives are: (1) to implement a YOLOv8-based computer vision pipeline that localizes pedestrians and identifies those crossing outside designated lanes, (2) to develop a logging system that timestamps and archives violation events for enforcement review, and (3) to quantitatively evaluate model performance using precision, recall, F1-score, and accuracy,

with statistical validation using a paired t-test to determine significant difference between the two models.

C. Significance

The system directly supports urban safety and efficient traffic management by automating jaywalking detection—reducing reliance on manual enforcement while providing consistent, data-driven monitoring. It offers the Local Government Unit (LGU) a scalable tool to improve pedestrian compliance and informs infrastructure planning through violation analytics, addressing critical safety needs highlighted by recent fatal incidents involving illegal crossings [9]. Additionally, the project serves as an applied AI case study for educational use and can be adapted by other Philippine cities facing similar pedestrian safety challenges.

D. Scope and Delimitations

This project is scoped to selected jaywalking-prone areas around Davao City. The model uses YOLOv8 to detect pedestrians and assesses violations based on foot position relative to the curb. It does not address physical infrastructure limitations such as absent sidewalks. Performance may be affected by occlusion, weather, lighting conditions, camera angle, and camera resolution. All video data are handled with limited retention in compliance with data privacy guidelines. Model evaluation uses a curated dataset of 453 images, with comparisons made between a pre-trained YOLOv8 model and a custom person detection model trained on over 5,000 images.

II. REVIEW OF RELATED LITERATURE

The integration of Artificial Intelligence (AI) and computer vision into traffic management systems has become a crucial advancement in promoting urban road safety. Studies have demonstrated that AI-based surveillance can effectively automate traffic rule enforcement, such as detecting violations and monitoring pedestrian behaviour, which traditionally relied on manual observation [10]. Deep learning, especially convolutional neural networks (CNNs), now delivers impressive accuracy for real-time object detection and tracking. Cities use these systems to spot road violations quickly and keep things running smoothly [11]. Pedestrian safety stands out as a real challenge in fast-growing cities like Davao City, where jaywalking keeps happening, no matter how many ordinances or public campaigns try to stop it. But with more CCTV cameras popping up across urban areas, there's a clear chance to use automated detection to help law enforcement and push for better compliance.

Recent research has emphasized the use of pre-trained object detection models such as the YOLO (You Only Look Once) family, and SSD for identifying human activities and traffic violations [12]. These models, when fine-tuned using domain-specific datasets, have achieved high precision and recall in complex environments, including crowded streets and varying lighting conditions [13]. YOLOv8, as the latest iteration, provides improved detection speed and generalization, making it suitable for real-time implementation in edge devices. Comparative evaluations in prior studies revealed YOLO's

superior inference efficiency over traditional R-CNN architectures, particularly for dynamic pedestrian detection [14]. Data augmentation and transfer learning have also been widely adopted to address limited datasets and enhance robustness under environmental constraints such as rain, low visibility, and occlusions; conditions often encountered in Philippine urban settings.

Several existing studies have specifically explored the use of AI for jaywalking detection. A 2022 study by Sharma et al. developed an intelligent system that localized jaywalking pedestrians in street videos using deep learning-based bounding box analysis and road segmentation [15]. Similarly, Bala and Kaushal proposed an improved localization method combining YOLOv5 with semantic segmentation to distinguish between lawful and unlawful crossings in real-time street footage [16]. In South Korea, Kim et al. implemented a vision-based jaywalking detection system leveraging YOLOv4 and DeepSORT to track pedestrians and determine violations under various lighting and traffic conditions [17]. These studies collectively demonstrate the growing global application of AI for detecting pedestrian violations, supporting the feasibility and significance of adopting a similar system for Davao City's traffic safety initiatives.

The rise of AI-assisted law enforcement systems throws ethical data handling and privacy protection into full view. These issues can't be ignored; not when technology steps into matters of justice and personal rights. To guarantee adherence to local data privacy regulations, a number of studies have combined face-blurring methods, anonymization procedures, and strict data retention guidelines [18]. Additionally, research on intelligent transportation systems indicate that predictive modelling and evidence-based policy-making are supported by the integration of detection models with geospatial data visualization platforms, such as dashboards or GIS-based tools [19]. These frameworks give authorities the ability to prioritize infrastructure upgrades or educational initiatives and pinpoint high-risk areas for jaywalking.

III. METHODOLOGY

A. Conceptual Framework

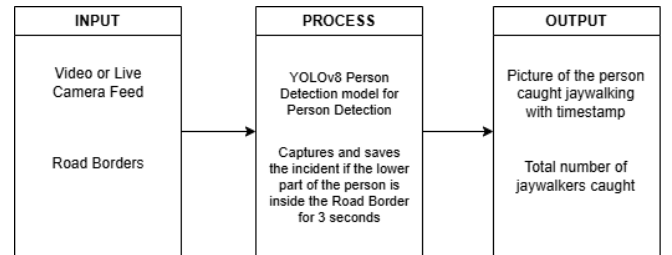


Fig. 1. Conceptual Framework of the System

Fig. 1 displays how the Jaywalking detection system works. The system requires a Live Camera Feed for its input since this will be used in security camera around the city. However, during the testing process, the developers used a video instead. Both formats are accepted as an input. Another input is the manual input of road borders which provide the Region of Interest (ROI) of the camera feed. Since the physical set-up stays in one

position only with the same angle, placing a static border would reduce power consumption and improve hardware performance. Thus, the developers did not use an instance segmentation model for road detection as it requires a higher quality of hardware material for its functionality and performance. Each input has its own role and its own process. For the road border, this serves as the detection area of the system. Thus, it must capture the road where the people jaywalk. The live feed input will be processed by inputting it on the person detection model. The model classifies the person and its lower part. The system checks for a specific condition. If the condition is met, then the system will provide an image output containing the jaywalker, at what position he or she has been caught, and the time stamp as the name format of the image.

B. System Flowchart

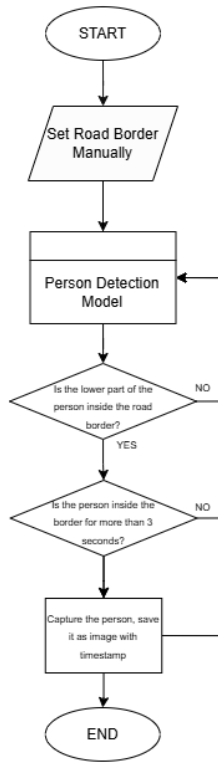


Fig. 2. Jaywalking Detection System Flowchart

Fig 2. above exhibits the main flow of the Jaywalking detection system. To start, the user must first create the road border by clicking at least the 3 points in the screen. Once the road has been selected as the Region of Interest (ROI), it will start using the Person Detection Model to observe and classify the people within the camera range. If the person detected stepped onto the road and 15% of the lower body is within the border, the system will start to count. If the timer reaches more than 3 seconds, the system will take a picture of the incident and label the person recorded as jaywalking. In addition, the time of capture is also recorded for future uses. If none of the conditions are met, the system will continue to observe the area until

someone performs jaywalking. The system will repeat until it is asked to be stopped.

C. Person Detection Development

The development of the model requires time to gather data, processing power for machine learning, and patience for testing and redeveloping the model. The developers found a person detection dataset which will be the inputs in creating their own Person Detection Model. This dataset contains 5,483 images and divides them by three: 4,407 images are used for training, 1,071 images for validation, and 5 images for testing [20]. This means that 80% of the images are used for testing while the remaining 20% is for validation and testing. The developers will use Pytorch in developing their own model having YOLOv8n as the model, 640 as the resolution size, and 80 for the number of epochs. The results and performance evaluation of the model created will be presented in the next chapter.

D. System Testing and Implementation

After creating the model, the developers will test the model's capability and efficiency by using video clips and compare its performance to an existing pre-trained model. The developers will gather video clips by capturing jaywalking incidents around the Davao City. The chosen locations are known for being a jaywalking hotspot since pedestrian lanes are far away and pedestrian would rather choose to jaywalk. The camera setup is on the sidewalk facing the road. It will be set-up at a right angle that will classify a person and placed a border in the roads. After gathering around 10 video clips, it will be fed in the system and check for the system's accuracy. This testing will serve as the basis whether the model created is efficient in fulfilling the objectives.

IV. RESULTS AND DISCUSSION

A. Model Training and Performance

The AI model was trained on 80 epochs to optimize its capability for detecting jaywalking violations. During the training process, key performance metrics were monitored to evaluate the model's learning progress. The training and validation losses, including bounding box loss, classification loss, and distribution focal loss, exhibited a consistent downward trend, indicating that the model was effectively learning to identify and localize relevant features from the data. Concurrently, core evaluation metrics such as precision, recall, and mean Average Precision (mAP) steadily increased, demonstrating the model's improving accuracy. The final model achieved a stable performance, with a peak mAP@.50 of approximately 0.88 and a mAP@.50-.95 of approximately 0.59, as visualized in the training performance graphs. This signifies that the model is highly proficient at correctly identifying jaywalking incidents with a high degree of confidence and localization accuracy.

B. Training Epoch Summary

To illustrate the model's progressive improvement, Table II presents a summary of key performance metrics at different intervals during the training phase. The data shows a clear

improvement from the initial epoch to the later stages of training, particularly in the mean Average Precision (mAP) scores, which are critical indicators of object detection performance.

TABLE II. PERSON DETECTION MODEL RESULTS

Epoch	Precision	Recall	mAP@.50	mAP@.50-.95
1	0.5462 ₃	0.53598	0.52066	0.24647
2	0.4961 ₂	0.49982	0.44517	0.19688
69	0.8411 ₈	0.72297	0.81048	0.52028
70	0.8443 ₅	0.71522	0.81062	0.52164
71	0.8581 ₈	0.70200	0.80990	0.52372

C. Comparison Results

To contextualize the performance of the proposed system, a comparative analysis was conducted against other established object detection models. Table III details the test results, comparing our model's accuracy with the pre-trained YOLOv8n model. This comparison highlights the effectiveness and competitive performance of our chosen AI-based approach for the specific task of detecting jaywalking violations in the context of Davao City.

TABLE III. TEST COMPARISON RESULTS

Test	Total Detected	Pre-trained Model Detected	Pre-trained Model Accuracy	Trained Model Detected	Trained Model Accuracy
1	6	6	100%	6	100%
2	1	1	100%	2	50%
3	3	3	100%	3	100%
4	2	6	33%	7	29%
5	2	3	66%	2	100%
6	2	2	100%	2	100%
7	6	3	50%	4	66%
8	2	1	50%	2	100%
9	9	6	66%	9	100%
10	4	4	100%	3	75%
11	5	2	40%	3	60%
12	6	6	100%	6	100%
13	5	5	100%	6	83%
14	3	3	100%	3	100%
15	1	1	100%	1	100%
16	1	1	100%	1	100%
17	2	1	50%	3	66%
18	2	2	100%	2	100%
TOTAL	60	56	80.83%	66	80%

As illustrated in Table III, the proposed model achieves a competitive balance when compared to a pre-trained YOLOv8n model. While the pre-trained model showed slightly higher precision, our model provides the optimal balance of precision and recall, which is critical for a practical application. High recall ensures that a majority of jaywalking incidents are successfully identified, while high precision minimizes false positives, ensuring that alerts or citations are accurate. This robust performance underscores the model's capability to

generalize and operate effectively in the challenging, dynamic urban environment of Davao City, validating its selection for this study.

V. CONCLUSION AND RECOMMENDATIONS

This study successfully developed an automated jaywalking detection system using computer vision and the YOLOv8 model to support traffic enforcement and enhance pedestrian safety in Davao City. By integrating real-time pedestrian detection with a rule-based decision component, the system can identify when pedestrians cross outside designated lanes and automatically capture evidence of the violation, including a timestamp image for visual proof. Model evaluation demonstrated strong performance, with the custom-trained YOLOv8 model achieving high precision, recall, and a mAP@.50 of approximately 0.88, indicating reliable detection. Comparative testing further showed that the trained model performs competitively against the pre-trained YOLOv8n model, particularly in challenging real-world video test scenarios. These results confirm that computer vision-based monitoring can reduce manual enforcement workload, provide continuous surveillance using existing CCTV infrastructure, and offer objective documentation of violations to support decision-making and urban safety initiatives. Ultimately, the system demonstrates the potential of providing AI-based solutions to contribute to a smarter traffic management across Philippine cities.

For future improvements, it is recommended to expand the dataset to include additional environmental variations such as heavy rain, nighttime conditions, and higher pedestrian density to further enhance model robustness. Integrating a tracking algorithm such as DeepSORT may also improve multi-pedestrian monitoring and reduce duplicate detections. Deployment on edge devices like NVIDIA Jetson, Raspberry Pi 5, or any similar hardware is suggested to enable real-time processing on-site without relying on centralized servers. Additionally, applying masking techniques such as face blurring can strengthen the system's data privacy regulations. Lastly, future studies may explore linking the system with enforcement databases or notification mechanisms for direct reporting and automated analytics dashboards to support city planning and pedestrian behaviour research.

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